

BLUESTOCK FINTECH

Mutual Fund Analytics Platform

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Final Project Report

1. Executive Summary

The Mutual Fund Analytics Platform was developed as part of the Bluestock Fintech Data Analyst Internship. The project focuses on collecting, preparing, storing, analysing, and presenting mutual fund and market-related data in a structured analytics workflow.

The project brings together information about mutual fund schemes, Net Asset Value (NAV), fund-house Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, investor transactions, portfolio holdings, and benchmark indices. These datasets are processed using Python and Pandas, stored in a SQLite database, and explored through SQL queries and business-intelligence dashboards.

The main purpose of the platform is to make mutual fund information easier to organise, compare, and understand. The analysis covers industry trends, fund performance, risk measures, investor transaction patterns, and comparisons with benchmark indices.

A reusable Python ETL script, `scripts/etl_pipeline.py`, was created and tested using a separate SQLite test database. The test load included 40 fund records, 1,297 date records, 46,000 NAV records, 32,778 transaction records, 40 performance records, and 90 AUM records.

During data preparation, an issue was identified in the cleaned NAV dataset. The cleaned file contained missing dates, so it was rebuilt using the original raw NAV file. The rebuilt file contained 46,000 rows, with no missing dates or duplicate scheme-date records.

The project provides a structured foundation for mutual fund analytics and dashboard-based exploration.

2. Data Sources

2.1 Overview

The project uses datasets relating to mutual fund schemes, market performance, industry trends, portfolio holdings, and investor transactions. The capstone brief identifies public information sources and supplied project datasets as inputs to the analytics workflow.

The datasets have different structures and reporting periods. They are prepared separately before being used for analysis and database loading.

2.2 Dataset Inventory

Dataset	Purpose
Fund master	Scheme details, fund house, category, plan, benchmark, and other metadata
NAV history	Daily NAV observations for selected schemes
AUM by fund house	Fund-house AUM observations
Monthly SIP inflows	Monthly SIP inflows and account-related measures
Category inflows	Net inflows grouped by category and month
Industry folio count	Industry folio totals and broad category breakdown
Scheme performance	Return and risk-related measures for selected schemes
Investor transactions	Synthetic transaction records with demographic and geographic fields
Portfolio holdings	Selected holdings, sectors, weights, and market values
Benchmark indices	Daily benchmark index closing values

2.3 Sources Named in the Capstone Brief

The capstone brief identifies the following sources:

- **AMFI India:** Mutual fund and industry-related information, including NAV, AUM, folio, and SIP information.
- **mfapi.in:** Historical NAV data accessed using scheme codes.
- **mfdata.in:** NAV and expense-ratio information.
- **NSE India and BSE India:** Benchmark index information.
- **AMFI Monthly Notes:** Monthly SIP and category-flow information.

2.4 Data Authenticity

The capstone brief describes scheme identifiers, scheme names, expense ratios, and AUM information as sourced from public information. NAV values are anchored to real historical values and simulated forward using realistic return and volatility parameters. Investor transaction records are synthetically generated for analytical practice.

This distinction is important when interpreting the results. Public-source information can be used to describe the corresponding source data, while synthetic transaction data is used to demonstrate analysis methods rather than to make claims about real customers.

3. ETL Design

3.1 Introduction

ETL stands for Extract, Transform, and Load. It is a process used to collect data from different sources, prepare it for analysis, and store it in a structured destination.

The project uses an ETL workflow to organise mutual fund datasets before they are used for SQL analysis and dashboard development.

3.2 ETL Architecture

The project follows the workflow:

Data Sources → Extract → Transform → Load → Analyse → Visualise

Extract

The extraction stage reads the available source datasets, including CSV files and public-source data identified in the capstone brief.

Each dataset contains information for a particular analytical purpose. For example, NAV history contains scheme-level daily values, while investor transactions contain transaction dates, amounts, scheme codes, and demographic fields.

Transform

The transformation stage prepares the data for reliable analysis. Python and Pandas are used to process the datasets.

The preparation work includes:

- Parsing date fields into a consistent format.
- Sorting NAV observations by scheme code and date.
- Checking duplicate records.
- Validating NAV values.
- Handling missing dates according to the cleaning rules.
- Standardising transaction types and categorical fields.
- Checking transaction amounts and KYC categories.
- Preparing numeric performance fields for analysis.

Load

The loading stage stores the prepared datasets in SQLite tables. SQLAlchemy and Pandas are used in the database-loading workflow.

A reusable Python script named `scripts/etl_pipeline.py` was created to load the prepared datasets into the database. The script was tested using a separate test database.

Analyse

After loading, the data can be explored using SQL queries and Python notebooks. The analysis includes data exploration, fund performance measures, investor transaction summaries, and benchmark comparisons.

Visualise

The final stage presents selected metrics and findings through interactive business-intelligence dashboards. The capstone brief specifies dashboard pages for industry overview, fund performance, investor analytics, and SIP and market trends.

3.3 NAV Data Preparation

The cleaned NAV dataset was checked for missing dates and duplicate scheme-date records.

During implementation, the cleaned NAV file was found to contain 27,842 missing dates. The original raw NAV file contained 46,000 rows with no missing dates. The cleaned file was therefore rebuilt from the original raw file.

The rebuilt file contained:

Validation	Result
NAV rows	46,000
Missing dates	0
Duplicate scheme-date records	0

This correction was important because missing or duplicate observations can affect time-based analysis and performance calculations.

3.4 ETL Test Results

The reusable ETL script was tested against a separate SQLite database. The following row counts were loaded during the test:

Table	Rows loaded
dim_fund	40
dim_date	1,297
fact_nav	46,000
fact_transactions	32,778
fact_performance	40
fact_aum	90

These results confirm that the listed datasets were loaded into the test database with the reported row counts. They do not, by themselves, confirm that every analytical query or dashboard measure is correct. Those require separate validation.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the mutual fund datasets and identify patterns in NAV, assets under management, SIP inflows, investor transactions, folio growth, and portfolio holdings. Charts and summary statistics were used to explore the data before conducting the detailed performance analysis.

4.1 Overview of EDA

The analysis uses cleaned mutual fund datasets, including NAV history, fund-house AUM, monthly SIP inflows, category-wise net inflows, investor transactions, folio counts, benchmark indices, and equity holdings.

The notebook uses visualisations such as line charts, bar charts, heatmaps, and pie charts to examine changes over time and differences between categories. The findings in this section describe the project dataset and should not be interpreted as statistics for the entire mutual fund industry unless independently verified.

4.2 NAV Trend Analysis

The NAV trend analysis examines the movement of Net Asset Value (NAV) over time. The EDA notebook includes NAV data for 40 mutual fund schemes and uses a line chart to explore changes in NAV across the available dates.

The NAV trend chart helps observe fluctuations and differences in the movement of the selected funds. The dashboard also includes a line chart titled “Selected Fund NAV vs NIFTY 100”, which displays the selected fund’s NAV alongside the NIFTY 100 index.

This chart provides a visual overview of the fund’s movement in relation to the benchmark. However, the two measures use separate vertical axes, so their line heights should not be compared directly to determine which performed better. A separate return calculation over the same period is required for a valid performance comparison.

The NAV trend chart is included in the Dashboard Screenshots section as Figure [6.2].

4.3 AUM Growth Analysis

Assets Under Management (AUM) represents the value of assets managed by a mutual fund or fund house. The AUM analysis compares changes in AUM across fund houses and over time.

The EDA notebook compares AUM across 10 fund houses for the period 2022–2025. In the recorded 2025 data, SBI Mutual Fund has an AUM of ₹12.50 lakh crore. The dashboard’s “AUM by Fund House” chart also places SBI Mutual Fund at the top of the displayed fund-house comparison.

The analysis helps show differences in the size of fund houses and the growth of AUM over the period covered by the dataset. These results describe the available project data and do not, by themselves, explain the reasons for AUM changes.

The AUM charts are included in the Dashboard Screenshots section as Figure [6.1].

4.4 SIP Inflow Analysis

Systematic Investment Plan (SIP) inflows represent the amount invested through SIPs during a given period. The notebook contains 48 monthly SIP records covering January 2022 to December 2025.

The notebook identifies December 2025 as the month with the highest recorded SIP inflow, at ₹31,002 crore. The monthly SIP chart helps observe changes in inflows over time, while the dashboard also compares SIP inflows with the NIFTY 50 index.

The chart provides a visual comparison of monthly SIP inflows and benchmark movements. Since these are different measures, the chart should not be interpreted as evidence that movements in the index caused changes in SIP inflows.

The SIP inflow chart is included in the Dashboard Screenshots section as Figure [6.4].

4.5 Investor Transaction Analysis

The investor transaction analysis examines transaction patterns by type, location, age group, and time. The notebook contains 32,778 investor transaction records.

The transaction-type analysis records:

- SIP: 19,716 transactions
- Lumpsum: 8,095 transactions
- Redemption: 4,967 transactions

SIP is the most frequent transaction type in the project dataset. The notebook also explores investor distribution by city tier, monthly transaction activity, and SIP investment amounts across states.

The dashboard provides additional comparisons through charts such as “Transaction Amount by State,” “Transaction Amount by Type,” and “Average SIP Amount by Age Group.” These visuals help examine differences in the recorded transactions. The results represent the project dataset, not verified investor behaviour across the entire industry.

The Investor Analytics dashboard screenshot is included in the Dashboard Screenshots section as Figure [6.3].

4.6 Category-wise Net Inflow Analysis

The category-wise net inflow analysis examines how net inflows vary across mutual fund categories and months. The notebook contains 144 category-inflow records, arranged across 12 categories and 12 months.

A heatmap is used to compare net inflows across categories and months. The dashboard also includes a “Top 5 Categories by Net Inflow (FY25)” chart, which allows categories with higher recorded net inflows to be identified.

The analysis shows that net inflows are not evenly distributed across the categories in the project dataset. The heatmap provides a broader view of monthly variation, while the top-five chart focuses on the categories with the largest displayed inflows.

The category-wise net inflow charts are included in the Dashboard Screenshots section as Figure [6.4].

4.7 Mutual Fund Folio Growth Analysis

A mutual fund folio is an account used to record an investor’s holdings in a mutual fund. Folio count analysis helps examine changes in the number of recorded investment accounts over time.

The notebook’s folio data shows that total folios increased from 13.26 crore in January 2022 to 26.12 crore in December 2025. This is an increase of approximately 97% over the period. The line chart shows an overall upward trend, with some smaller changes between recorded periods.

The analysis indicates growth in the total folio count within the dataset. However, folio count represents accounts and should not be treated as the number of unique investors.

4.8 NAV Return Correlation Analysis

Correlation analysis was performed to examine how the daily returns of selected mutual fund schemes moved in relation to one another. The notebook calculates a correlation matrix for 10 selected funds using their daily NAV returns.

The displayed correlation values range from -0.39 to 0.68 . The positive values indicate that some selected funds’ daily returns tended to move in the same direction, while the negative values indicate that some tended to move in opposite directions. The strength of this relationship varies between fund pairs.

The correlation matrix helps identify similarities and differences in the return movements of the selected funds. Correlation measures co-movement and does not establish that one fund’s movement caused another’s.

4.9 Sector Allocation Analysis

The sector allocation analysis examines the distribution of equity holdings across sectors in the holdings dataset. The notebook contains 322 holding records and groups the recorded holding weights by sector.

Banking has the largest summed holding weight in the notebook's sector aggregation, followed by IT and Pharma. The analysis provides an overview of which sectors account for larger combined weights across the holdings records.

These figures are sums across the available holdings records. They should not be interpreted as percentages of one portfolio or as the allocation of the entire mutual fund industry. The analysis is limited to the holdings included in the project dataset.

4.10 Summary of EDA Findings

The EDA provided an overview of the project data through trend analysis, category comparisons, and investor transaction analysis. The main observations were:

1. NAV values varied over time across the selected mutual fund schemes.
2. SBI Mutual Fund had the highest displayed AUM among the fund houses in the notebook's 2025 comparison.
3. December 2025 had the highest recorded monthly SIP inflow in the notebook.
4. SIP was the most frequent transaction type in the investor transaction dataset.
5. Total folios increased from 13.26 crore in January 2022 to 26.12 crore in December 2025.
6. Daily return correlations differed across the 10 selected funds.
7. Banking had the largest summed sector weight in the available holdings records.

These findings provide a starting point for the detailed performance analysis in the next section.

5. Performance Analysis

5.1 Overview of Performance Analysis

Performance analysis was carried out to evaluate the returns and risk characteristics of the mutual fund schemes included in the project. The analysis used historical NAV data and benchmark index data to calculate and compare different performance measures.

The measures included Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. A weighted Fund Scorecard was also created to rank the schemes using multiple performance and risk measures. Finally, the top five funds were selected for comparison with the NIFTY 50 and NIFTY 100 benchmarks over a three-year period.

These calculations help provide a broader view of fund performance rather than relying on returns alone.

5.2 Compound Annual Growth Rate (CAGR) Analysis

Compound Annual Growth Rate (CAGR) measures the annualised rate at which an investment grows over a specified period. In this project, CAGR was calculated for 1-year, 3-year, and 5-year periods using the historical NAV data of each fund.

For each scheme, the notebook identified the NAV closest to the target starting date and compared it with the latest available NAV for that scheme. The CAGR calculation was then applied to estimate the annualised growth rate over the selected period.

The CAGR results were arranged by fund code and scheme name, allowing the returns of different schemes to be compared. The analysis provides a way to examine fund growth over different investment horizons.

5.3 Sharpe Ratio Analysis

The Sharpe Ratio was calculated to examine the return earned by a fund in relation to its volatility. The notebook used a risk-free rate of 6.5% and 252 trading days for annualisation.

Daily returns were used to calculate the annualised average return and annualised standard deviation. The Sharpe Ratio was then calculated by comparing the annualised return above the risk-free rate with annualised volatility.

The schemes were ranked according to their Sharpe Ratios. A higher Sharpe Ratio indicates a higher return relative to the measured volatility under the assumptions used in this project.

The Sharpe Ratio provides additional information about risk-adjusted performance, which cannot be understood from CAGR alone.

5.4 Sortino Ratio Analysis

The Sortino Ratio was calculated to assess fund returns in relation to downside risk. Unlike the Sharpe Ratio, which uses overall return volatility, the Sortino Ratio uses the standard deviation of negative daily returns.

The notebook identified negative daily returns for each fund and calculated downside standard deviation. It then compared the annualised return above the 6.5% risk-free rate with the annualised downside deviation.

The funds were ranked according to their Sortino Ratios. This measure helps distinguish funds that experienced less downside volatility from those with greater downside risk.

The Sortino Ratio complements the Sharpe Ratio by focusing specifically on negative return movements.

5.5 Alpha and Beta Analysis

Alpha and Beta were calculated by comparing the daily returns of each mutual fund with the daily returns of the NIFTY 100 index. The benchmark data was sorted by date, and daily benchmark returns were calculated from changes in the index closing values.

Fund returns were then matched with NIFTY 100 returns using their common dates. A linear regression was performed for each fund, with the fund's daily return as the dependent variable and the benchmark's daily return as the independent variable.

Beta represents the estimated sensitivity of the fund's returns to movements in the NIFTY 100. A higher Beta indicates greater sensitivity to benchmark movements, while a lower Beta indicates less sensitivity.

Alpha was calculated from the regression intercept and annualised using 252 trading days. It represents the estimated return component not explained by the benchmark relationship under the model used.

These measures help examine how the funds' daily returns relate to the NIFTY 100 benchmark. They should be interpreted as estimates based on the selected benchmark and the period covered by the data.

5.6 Maximum Drawdown Analysis

Maximum Drawdown was calculated to measure the largest decline in NAV from a previous peak during the available observation period.

For each fund, the notebook sorted the NAV records by date and calculated the running maximum NAV. Drawdown was then calculated by comparing each NAV value with its running maximum. The lowest drawdown value was recorded as the fund's Maximum Drawdown.

The notebook also identified the date of the maximum drawdown and the preceding peak date. Maximum Drawdown was calculated for all 40 funds.

This measure helps describe the largest observed decline from a previous NAV peak. It provides information about the historical downside experience of a fund, but it does not predict the size of future losses.

5.7 Fund Scorecard and Ranking

A weighted Fund Scorecard was created to compare the 40 mutual fund schemes using multiple measures. The scorecard combined 3-year CAGR, Sharpe Ratio, Alpha, expense ratio, and Maximum Drawdown.

Each measure was ranked separately. Funds with higher CAGR, Sharpe Ratio, and Alpha received better ranks. A lower expense ratio was treated as preferable, while a less negative Maximum Drawdown was considered better.

The ranks were converted into scores on a 0–100 scale and combined using the following weights:

Performance measure	Weight
3-year CAGR	30%
Sharpe Ratio	25%
Alpha	20%
Expense Ratio	15%
Maximum Drawdown	10%

The final Fund Score was used to rank the schemes. This approach provides a combined comparison of return, risk, and cost-related measures instead of evaluating each fund using a single metric.

The Fund Score is a project-specific ranking based on the selected measures and weights. It should not be treated as a guarantee of future performance or as an investment recommendation.

5.8 Benchmark Comparison

The top five funds from the Fund Scorecard were selected for comparison with the NIFTY 50 and NIFTY 100 indices over a three-year period.

The comparison period recorded in the notebook is from 29 May 2023 to 29 May 2026. The NAV values of the selected funds and the closing values of the benchmark indices were normalised to a starting value of 100. This allowed their relative movement over the comparison period to be displayed on a common starting basis.

A line chart titled **“Top 5 Funds vs NIFTY 50 and NIFTY 100 — 3-Year Performance”** was created to display the normalised values over time. The notebook also calculated tracking error for each of the top five funds against both benchmark indices.

Tracking error measures the variability of the difference between a fund’s daily returns and the benchmark’s daily returns. It provides information about how closely the fund’s return movements followed the selected benchmark during the comparison period.

The benchmark comparison supports a visual examination of relative movement. However, the final performance conclusion should be based on the calculated return values over the same period, rather than comparing line heights alone.

5.9 Summary of Performance Analysis

The performance analysis evaluated the mutual fund schemes using return, risk, and benchmark-related measures. CAGR was used to examine annualised growth, while the Sharpe and Sortino Ratios were used to assess returns in relation to volatility and downside risk. Alpha and Beta were calculated to examine the relationship between fund returns and the NIFTY 100 index. Maximum Drawdown was used to measure the largest observed decline from a previous NAV peak.

The weighted Fund Scorecard combined selected measures to rank the schemes, and the top five funds were selected for a three-year comparison with the NIFTY 50 and NIFTY 100 benchmarks.

Together, these measures provide a more complete view of the historical performance and risk characteristics of the funds included in the project dataset.

6. Dashboard Development and Insights

6.1 Overview

An interactive dashboard was developed using Power BI to present the key findings from the mutual fund analytics project. The dashboard brings together information about the mutual fund industry, fund performance, investor transactions, SIP activity, and market trends.

The dashboard is divided into four pages:

1. Industry Overview
2. Fund Performance
3. Investor Analytics
4. SIP & Market Trends

Each page contains visualizations and key performance indicators (KPIs) to help users explore the data and understand important trends. Filters and slicers allow users to view information for selected years, states, funds, or other available categories.

6.2 Industry Overview

The Industry Overview page presents a summary of the mutual fund industry's assets under management (AUM), SIP activity, and folio growth.

The page includes the following visualizations:

- **Industry AUM Trend:** Shows the change in total AUM over time.
- **AUM by Fund House:** Compares AUM across different fund houses.
- **Key Performance Indicators:** Displays total AUM, SIP inflow, and total folios.

The dashboard shows an overall increase in AUM across the displayed period. The fund-house comparison also highlights differences in the AUM managed by individual fund houses.

These visualizations provide a quick overview of the industry's growth and help users compare fund houses.

6.3 Fund Performance

The Fund Performance page focuses on the movement of mutual fund NAV and its comparison with market benchmarks.

The page includes a chart titled “**Selected Fund NAV vs NIFTY 100.**” It allows users to examine the movement of a selected fund's NAV alongside the NIFTY 100 index over time.

The page also supports the analysis of fund performance through the metrics calculated in the performance analytics stage, including CAGR, Sharpe Ratio, Sortino Ratio, alpha, beta, and maximum drawdown.

The comparison helps users explore how the selected fund has moved in relation to the benchmark. However, NAV and index closing values use different scales, so their raw line heights should not be compared directly. The chart is intended to show movement over time rather than a direct comparison of the two raw values.

6.4 Investor Analytics

The Investor Analytics page presents information about investor transactions and their distribution across different categories.

The page includes:

- **Transaction Amount by State:** Compares transaction amounts across states.

- **Transaction Type Distribution:** Shows the distribution of transaction amounts by transaction type.
- **Average SIP by Age Group:** Compares average SIP amounts across age groups.
- **Monthly Transaction Volume:** Displays changes in transaction activity over time.

These visualizations help explore differences in transaction activity across locations, transaction types, and age groups.

The transaction type chart represents the distribution of transaction amounts, while the transaction records analysis also includes transaction counts. These measures should be interpreted separately because the number of transactions and the total amount transacted are different metrics.

6.5 SIP and Market Trends

The SIP & Market Trends page presents trends in SIP inflows and compares them with market movement.

The page includes the following visualizations:

- **SIP Inflows vs NIFTY 50:** Shows monthly SIP inflows alongside NIFTY 50 movement.
- **Category Inflow Heatmap:** Displays category-wise inflow patterns over time.
- **Top Categories:** Compares categories based on the inflow measure used in the dashboard.

The SIP inflow chart provides a view of changes in monthly SIP activity. The category heatmap helps identify differences in inflows across categories and periods.

SIP inflows and index values are different measures and use different scales. Their combined chart should therefore be interpreted as a comparison of trends, not as a direct comparison of raw values.

6.6 Dashboard Interactivity

The dashboard includes slicers and filters to support interactive exploration. Users can select available values such as year, state, fund, or transaction category to narrow the displayed information.

This allows users to explore specific parts of the dataset without creating separate charts for every possible selection.

The dashboard is designed to make the results easier to understand by combining summary indicators with detailed visualizations.

6.7 Dashboard Screenshots

Screenshots of the four dashboard pages are included below to demonstrate the final dashboard design and its visualizations.

Figure 6.1: Industry Overview Dashboard



Figure 6.2: Fund Performance Dashboard



Figure 6.3: Investor Analytics Dashboard



Figure 6.4: SIP & Market Trends Dashboard



6.8 Dashboard Limitations

The dashboard is based on the datasets prepared for this project. Some data, including investor transaction records, is simulated. Therefore, the dashboard findings should be interpreted as results from the project dataset and not as verified statistics about all mutual fund investors or the entire industry.

Some charts combine measures that use different scales. These charts are useful for exploring movement and patterns, but the raw values should not be compared directly.

The dashboard also depends on the accuracy and completeness of the underlying data. Any missing values, unusual records, or differences in units may affect the displayed results.

6.9 Summary

The Power BI dashboard provides an interactive view of mutual fund industry trends, fund performance, investor transactions, and SIP activity. The four dashboard pages organize the analysis into clear sections and allow users to explore the prepared dataset using filters and visualizations.

The dashboard supports the project's objective of using data analytics to present mutual fund information in a clear and accessible format.

7. Key Findings and Recommendations

7.1 Overview

The mutual fund analytics project involved data preparation, ETL development, database creation, exploratory data analysis, performance and risk analysis, and dashboard development.

The project brought together different datasets to explore mutual fund schemes, industry trends, investor transactions, SIP activity, and market performance. The following findings and recommendations summarize the overall work completed.

7.2 Key Findings

7.2.1 Data Preparation and ETL

The project demonstrated the importance of preparing data before performing analysis. The datasets required checks for missing values, duplicate records, date formats, and numerical values.

An ETL pipeline was developed to organize the data-loading process. The cleaned data was then loaded into a SQLite database using a structured schema.

This process helped organize the datasets and made them available for further analysis using SQL and Python.

7.2.2 Database and SQL Analysis

A SQLite database was created using a star-schema approach. The database included dimension and fact tables for mutual fund schemes, dates, NAV, transactions, performance, and AUM.

SQL queries were used to explore the data and answer analytical questions related to fund assets, NAV trends, SIP activity, transactions, and scheme performance.

The database structure provided a consistent way to store and query information from multiple datasets.

7.2.3 Exploratory Data Analysis

Exploratory data analysis was used to understand the structure and patterns in the prepared datasets.

The analysis covered:

- NAV movement across mutual fund schemes.
- AUM trends and comparisons across fund houses.
- Monthly SIP inflows and category-wise inflows.
- Investor transaction activity by type, location, and age group.
- Folio growth and sector-wise holdings.
- Relationships between selected fund NAV returns.

These analyses helped identify patterns within the project dataset and provided a foundation for the performance analysis and dashboard.

7.2.4 Fund Performance and Risk Analysis

The performance analysis calculated CAGR, Sharpe Ratio, Sortino Ratio, alpha, beta, and maximum drawdown for the selected schemes.

A weighted fund scorecard was also created to compare schemes using performance and risk-related measures. The scorecard combined the selected metrics into a comparative ranking.

The analysis provided a structured way to examine the schemes, but the results depend on the available data and the methodology used. In particular, a full five-year CAGR should only be interpreted as valid when sufficient five-year NAV history is available.

7.2.5 Investor and SIP Analysis

The investor analysis explored transaction activity across transaction types, states, age groups, and months. SIP analysis examined monthly inflows and differences across categories.

These analyses demonstrated how transaction and SIP datasets can be used to explore investor activity. However, because the investor transaction records are simulated, the findings describe the project dataset and should not be treated as verified conclusions about actual investor behaviour.

7.2.6 Dashboard Development

The Power BI dashboard brought together the project's main analytical areas in four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends.

The dashboard used KPIs, charts, and filters to make the prepared data easier to explore. It provided a visual summary of the analysis and allowed users to focus on selected funds, periods, states, and categories.

7.3 Recommendations

7.3.1 Strengthen Data Quality Checks

Data validation should remain an important part of the analytics process. Future improvements could include more automated checks for missing values, duplicate records, invalid dates, unexpected values, and inconsistent units.

7.3.2 Improve Historical Data Coverage

A longer period of verified NAV data would improve the reliability of long-term performance calculations. Performance metrics should be calculated only when the required historical data is available.

7.3.3 Validate ETL and Database Results

The ETL pipeline should continue to include row-count checks and validation after loading data into the database. This would help identify missing or duplicated records and ensure that the database reflects the cleaned datasets.

7.3.4 Improve Performance and Benchmark Comparisons

Fund and benchmark comparisons should use clearly defined periods and matching dates. The analysis should also clearly distinguish between fund NAV and benchmark index values, as they represent different measures.

7.3.5 Use Reliable Data for Investor Analysis

The simulated transaction dataset is useful for demonstrating analytical methods. Further conclusions about real investor behaviour would require reliable and representative transaction data.

7.3.6 Improve Dashboard Clarity

Dashboard titles, labels, units, and filters should be reviewed for consistency. Charts that display measures with different scales should clearly communicate this so that users do not compare raw values incorrectly.

7.3.7 Extend the Project in Future

Future improvements could include additional verified historical data, further validation of risk metrics, more detailed investor cohort analysis, and additional dashboard features. These improvements could provide deeper insights while maintaining clear documentation of the data and methodology.

7.4 Project Limitations

The project has certain limitations that should be considered when interpreting the findings.

- Some datasets, including investor transaction records, are simulated or prepared for analytical purposes.
- Some NAV values were simulated to extend the available history.
- The available NAV history may not support a valid full five-year performance calculation for every scheme.
- Performance results depend on the selected formulas, benchmark, time period, and data quality.
- Dashboard findings reflect the prepared project datasets and are not necessarily representative of the entire mutual fund industry.

These limitations do not reduce the value of the project as an analytics exercise, but they are important when interpreting its results.

7.5 Summary

The project demonstrated an end-to-end mutual fund analytics workflow, covering data preparation, ETL, database development, SQL analysis, exploratory data analysis, performance and risk analysis, and dashboard development.

The work shows how multiple datasets can be organized and analyzed to explore mutual fund trends and performance. It also highlights the importance of data quality, consistent methodology, and careful interpretation.

The project provides a foundation for further analysis using more complete and verified data.

8. Conclusion

This project provided practical experience in applying data analytics to mutual fund data. It covered data preparation, ETL pipeline development, database creation using SQLite, SQL-based analysis, exploratory data analysis, fund performance and risk analysis, and dashboard development using Power BI.

The project helped demonstrate how data from different sources can be organized and analyzed to understand mutual fund trends, scheme performance, investor transactions, and SIP activity. The interactive dashboard presented the main findings in a clear and accessible format.

Overall, the project strengthened my understanding of the end-to-end data analytics process and the use of tools such as Python, SQL, SQLite, and Power BI. It also highlighted the importance of data quality, appropriate analytical methods, and careful interpretation of results.

The project serves as a foundation for further learning and for future analytics projects using more complete and verified data.